

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages & DSA:** Python, SQL, Data Structures & Algorithms (Arrays, Hashing, Trees, Graphs, Dynamic Programming), LeetCode
• **Backend & Systems:** FastAPI, REST APIs, Microservices, Distributed Systems, ETL Pipelines, Kafka, PySpark
• **Databases, Cloud & Tools:** PostgreSQL, Neo4j, AWS (S3, EC2, EKS), GCP, Docker, Git, Prompt Engineering, AWS Bedrock, AWS Lambda, AWS API Gateway, GitHub Copilot

WORK EXPERIENCE

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

• Built and deployed scalable RAG system with structured content indexing and vector embeddings using FastAPI and PostgreSQL, enabling AI-assisted content discovery and retrieval in production
• Designed efficient, fault-tolerant ETL and event-driven async pipelines to process and index structured document datasets for low-latency AI-assisted retrieval
• Improved system performance by **64%** through optimized indexing strategies and pipeline improvements in production environments
• Developed backend services enabling real-time, high-performance **AI model serving** of unstructured data across distributed systems

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

• Owned **end-to-end design, development, and integration** of cross-platform **AI backend system** ingesting wearable data for 100+ users, enabling scalable data collection and downstream analytics
• Designed and optimized **PostgreSQL schema, indexing strategies, and distributed query execution plans** for high-volume datasets, improving query performance by 38% in production workloads
• Built production-ready Python generative AI pipeline with prompt engineering, retrieval, and context summarization, improving response accuracy by 43%
• Developed serverless backend workflows using AWS Lambda for real-time, fault-tolerant event-driven data processing with focus on scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Designed and deployed highly **scalable NLP and generative AI pipelines** in **Python** to process datasets, reducing processing latency by **41%**
• Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing
• Designed and developed production-grade REST APIs using FastAPI and AWS API Gateway to serve AI models, implementing validation, monitoring, and deployment automation across distributed systems
• Experimented with AI-assisted development tools and prompt optimization techniques to engineer workflows and improve system efficiency

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

• Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving operational visibility
• Processed and optimized large-scale datasets using **Python** and **PySpark** to identify failure and performance issues in distributed systems
• Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data issues, speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

• Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, leveraging **SQL** and **DAX**, enabling KPI tracking and data-driven decisions
• Automated recurring **Excel**-based workflows using **VBA**, implementing validation and scheduling logic, reducing report preparation time by **42%** and improving consistency

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

• Led cross-functional team of **6** engineers and designers to deliver **AI-driven product features** for DeepCover game, leveraging feedback and performance insights for feature prioritization
• Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, managing backlog and issue resolution

PROJECTS

Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

• Built end-to-end ML system using Python, Pandas, and LightGBM to assess credit risk from time-series transaction data, covering data preprocessing, feature engineering, model training, and evaluation
• Designed and implemented **scalable ETL and feature engineering pipelines**, capturing behavioral trends and temporal patterns such as rolling averages and spending frequency to improve model performance
• Developed a **decision framework for credit line adjustments**, incorporating model outputs and risk thresholds to improve prediction precision by 20% and enable risk-aware classification of customers